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Figure 1: Yosemite Valley from Wawona Road (Highway 41), October 10, 2022. El Capitan is visible on the left, Half Dome near the center, and Cathedral Rocks and Spires on the right. Photograph by Yoni Rodriguez.

SHM 377: Hazardous Materials Management

What Is a Hazardous Material?

Lecture 01 · Fall 2026

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What Counts as a Hazardous Material?

Before we define a hazardous material, let's see what you think.

Poll Everywhere

Which of the following would you consider a hazardous material?

Select all that apply.

- Gasoline
- Lead
- Asbestos
- Pesticides
- Lithium-ion batteries
- Smoke
- Ionizing radiation
- Non-ionizing radiation
- Heat
- Cold
- Bacteria
- Viruses

Use the same Poll Everywhere response page from the beginning of class.



What Made You Choose?

Why did you consider some of these hazardous materials?

Why might you have hesitated on others?

What made something seem hazardous?

- Could it make someone sick?
- Could it cause an injury?
- Could it burn or explode?

- Could it damage property?
- Could it damage the environment?
- Could someone be exposed to it?

These are all useful questions.

But there is another question we need to answer:

Is everything that is hazardous a hazardous material?

Types of Hazards

Hazards can take many forms.

Chemical

- Toxic substances
- Flammable materials
- Corrosives
- Reactive materials
- Explosives



Figure 2: West Fertilizer Company explosion, West, Texas, April 17, 2013. Credit: U.S. Chemical Safety and Hazard Investigation Board.

Physical

- Heat
- Cold
- Noise
- Ionizing radiation
- Non-ionizing radiation
- Ergonomic stressors

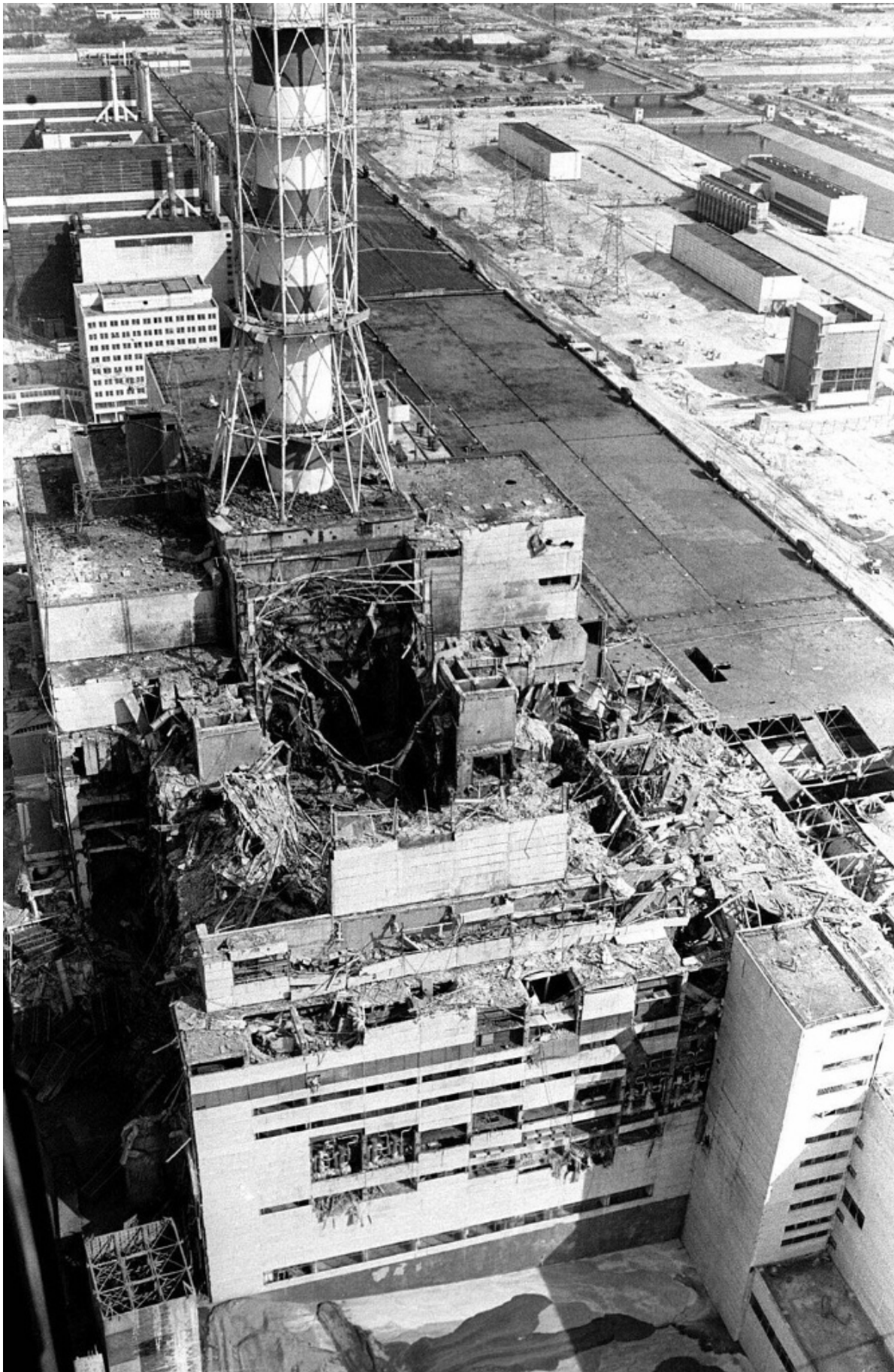


Figure 3: Chernobyl Nuclear Power Plant following the April 26, 1986 reactor accident. Photo: Igor Kostin / Novosti Press Agency.

Biological

- Bacteria
- Viruses
- Mold
- Vector-borne hazards



Figure 4: Health workers disinfecting a morgue and preparing to bury a probable Ebola victim in Rwampara, Democratic Republic of the Congo, September 2026. Photo: Arlette Bashizi / The New York Times.

All of these can cause harm.

But **not every hazard is a hazardous material.**

Hazard Hazardous Material

A **hazard** is something with the potential to cause harm.

A **hazardous material** is a material or substance whose properties can pose a risk to:

- human health
- safety
- property
- the environment

Consider some of our Poll Everywhere examples.

Heat can be hazardous, but heat is not a material.

Cold can be hazardous, but cold is not a material.

Noise can damage hearing, but noise is not a material.

Ionizing radiation can cause harm, but radiation is energy rather than matter.

A **radioactive material**, however, contains unstable atoms that emit ionizing radiation.

All hazardous materials present hazards, but not all hazards are hazardous materials.

Where Does SHM 377 Fit?

This course focuses on **hazardous materials and their management**.

But hazardous-material disasters can involve several interacting factors.

Deepwater Horizon

Physical hazards

- **Pressure Differential:** “The sudden, massive drop in hydrostatic pressure allowed trapped gas to expand violently as it traveled upward.” [Source: EBSCO Research Starters](#)
- **Equipment Overwhelm:** “The rig’s mud-gas separator was physically incapable of handling the volume and speed of the rushing blowout. It failed, raining highly flammable drilling fluids and gas directly onto the rig deck.” [Source: U.S. Department of the Interior](#)

Chemical hazards

- **Flammable Gas:** Methane
- **Ignition Source:** “The expanding vapor cloud quickly encountered an ignition source on the rig (likely from the electrical system or engine rooms). This caused a rapid chemical combustion reaction, resulting in a massive fireball explosion at 9:49 p.m., followed by a secondary explosion moments later.” [Source: U.S. Department of the Interior](#)

Human and organizational decisions

- Misinterpreted pressure-test results
- Failure to recognize warning signs
- Decisions that reduced safety margins
- Inadequate response to the developing well-control problem

The disaster resulted from these factors **acting together**, not from a single hazard alone.



Figure 5: Deepwater Horizon offshore drilling rig fire, Gulf of Mexico, April 21, 2010. Photo: Gerald Herbert / Associated Press.

What Is a Hazardous Material?

At a broad level, a hazardous material is a substance or material capable of causing harm to:

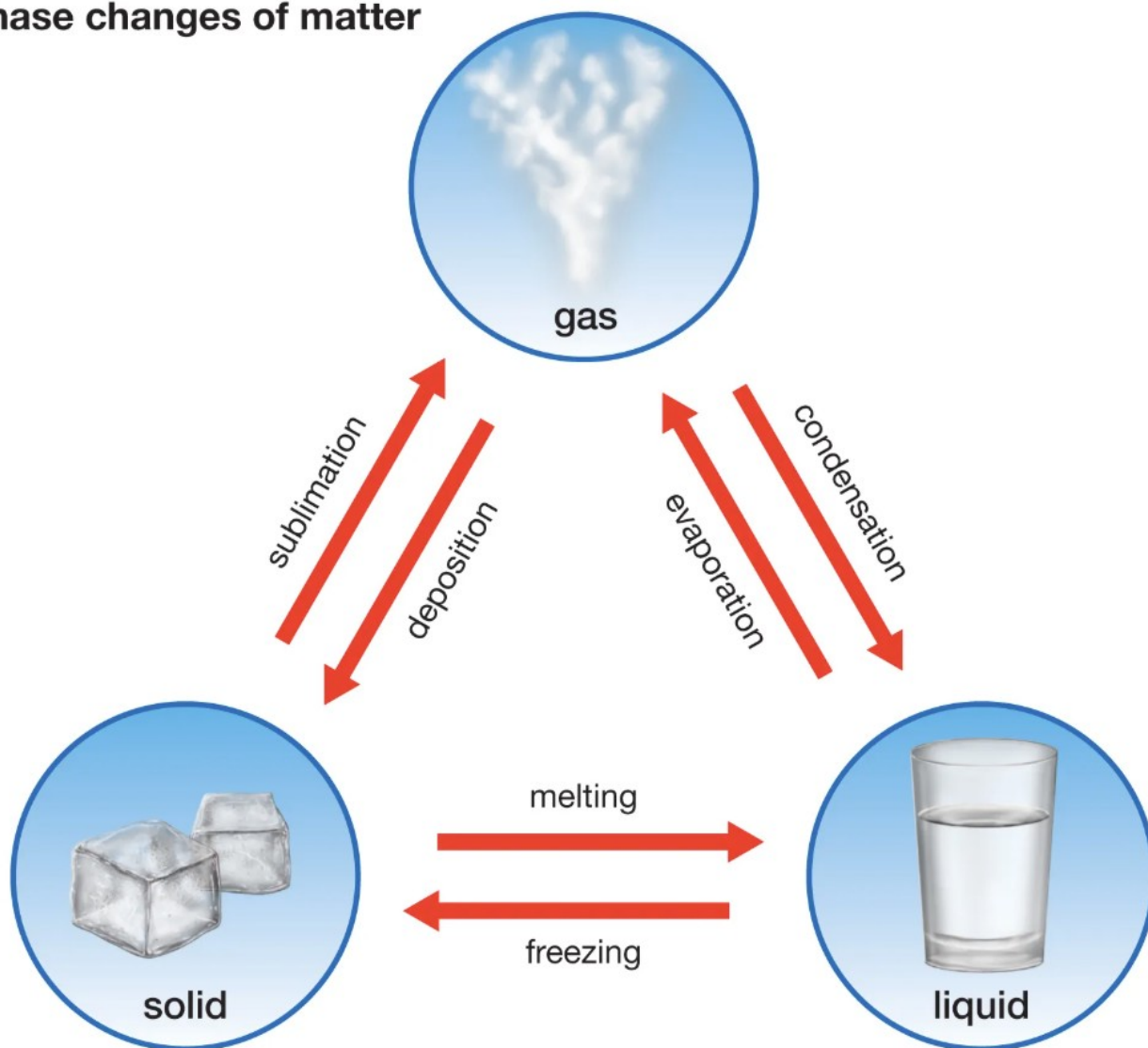
- human health
- safety
- property
- the environment

Hazardous materials may exist as:

- **solids**
- **liquids**
- **gases**

What makes them hazardous depends on their physical, chemical, biological, or radioactive properties.

Phase changes of matter



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Figure 6: Phase changes of matter between solids, liquids, and gases.

Hazardous Properties

Some important hazardous properties include:

- **Flammability:** ability to ignite and burn
- **Corrosivity:** ability to damage living tissue or materials
- **Toxicity:** ability to cause harmful health effects
- **Reactivity:** ability to undergo dangerous chemical reactions
- **Explosivity:** ability to rapidly release energy and pressure
- **Oxidizing potential:** ability to intensify combustion
- **Pressure:** hazards associated with compressed or liquefied gases
- **Radioactivity:** emission of ionizing radiation
- **Biological hazard:** presence of disease-causing organisms or biological agents

A material may have **more than one hazardous property**.

Does the Presence of a Hazard Mean You Will Be Harmed?

Not necessarily.

Knowing that something **has a hazardous property** is only the beginning.

To understand the potential for harm, we also need to ask:

- How much are you exposed to?
- How often are you exposed?
- How long does the exposure last?
- How does it enter or interact with the body?
- What biological effect can occur at that dose?

This relationship between the amount of exposure and the resulting effect is called the **dose-response relationship**.

“Only the Dose Makes the Poison”



Figure 7: Portrait of Paracelsus, attributed to Quentin Matsys or his workshop/circle, c. 1530–1540.

Paracelsus (1494–1541) was a Swiss physician and alchemist often called the **Father of Toxicology**. He is associated with one of the foundational principles of toxicology:

“Only the dose makes the poison.”

The important question is therefore not simply:

Is this hazardous?

It is also:

At what dose does it become harmful?

From Hazard to Health Effect

The presence of a hazard does not necessarily mean that harm will occur.

Hazardous Property

↓

Exposure

↓

Dose

↓

Biological Response

The **dose** helps connect exposure to the potential health effect.

Two people can encounter the same hazardous material but receive very different doses depending on the:

- concentration or intensity
- frequency of exposure
- duration of exposure
- route of exposure

Bananas Are Radioactive

Bananas naturally contain **potassium**.

A small fraction of naturally occurring potassium is the radioactive isotope **potassium-40**.

So technically:

Bananas are radioactive.

But does that mean you should stop eating bananas?



Figure 8: Bananas with a radiation warning symbol. Photo: Stepan Popov.

What Do You Think?

Poll Everywhere

Before we look at the dose, make your best guess.

1. Should You Stop Eating Bananas?

- Yes
- No
- Not sure

2. Approximately how many bananas would correspond to a potentially lethal radiation dose?

- 10
- 100
- 1,000
- 100,000
- 1,000,000
- 50,000,000



Radioactive Automatically Dangerous

The important question is not simply:

Is radiation present?

The important question is:

What dose does a person receive?

A banana contains a very small amount of naturally occurring radioactive potassium-40.

Eating a banana therefore exposes you to a **very small amount of ionizing radiation**.

The presence of radioactivity alone does not tell us whether the exposure represents a meaningful health risk.

How Much Radiation Are We Talking About?

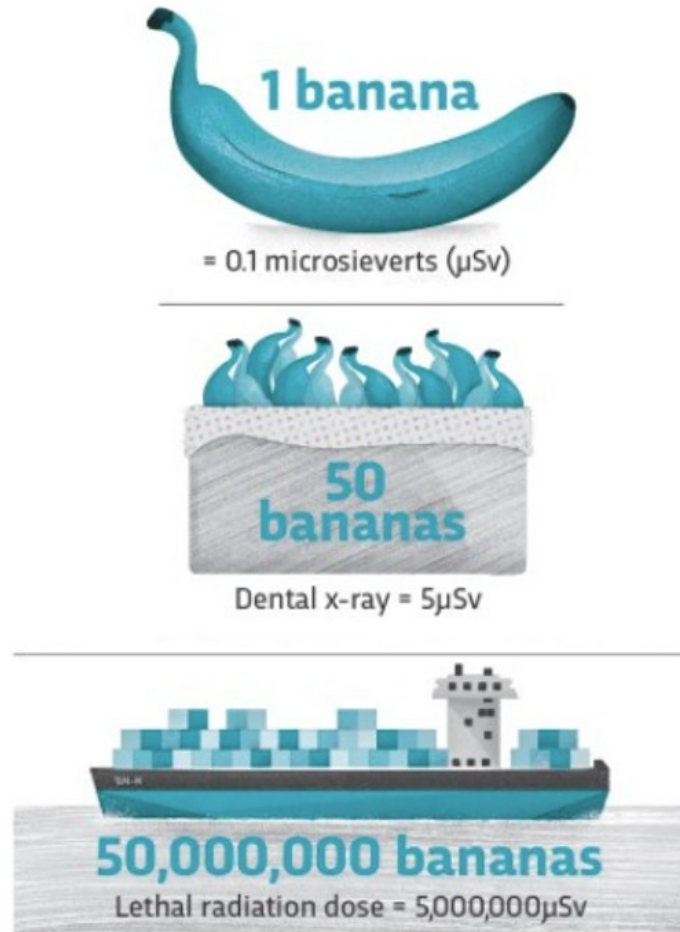


Figure 9: Illustration comparing radiation doses using banana equivalent doses. Credit: Information is Beautiful / David McCandless.

The **banana equivalent dose** is a communication tool used to put very small radiation doses into perspective.

The point is **not** that someone could realistically eat millions of bananas.

The point is that:

Detectable radiation is not the same thing as a dangerous radiation dose.

Hazard, Exposure, Dose, and Response

This gives us a more complete way to think about hazardous materials:

Hazard

↓

Exposure

↓

Dose

↓

Response

A material can possess a **hazardous property** without every encounter with that material producing harm.

Understanding risk requires us to consider both:

What can the hazard do?

and

How much exposure actually occurs?

Does Everyone Define “Hazardous Material” the Same Way?

Not necessarily.

Different organizations define and classify hazardous materials based on the problem they are trying to manage.

Workplace

OSHA / Washington L&I

How can workers be protected from hazardous chemicals and exposures?

Transportation

U.S. Department of Transportation

Can the material pose an unreasonable risk during transportation?

Environment & Waste

EPA / Washington State Department of Ecology

How should releases, contamination, and hazardous wastes be managed?

Emergency Response

NFPA and other emergency-response systems

What hazards do responders need to recognize quickly?

The same material may therefore be described or regulated differently depending on **where it is, how it is being used, and what is happening to it.**

Hazard Communication Is Part of Management

Throughout SHM 377, we will encounter several systems used to communicate information about hazardous materials.

- GHS / OSHA labels and Safety Data Sheets
- DOT hazard classes and placards
- NFPA 704 hazard identification
- Environmental and hazardous-waste classifications

You will learn how to recognize and use these systems.

But hazardous-material management requires more than recognizing a symbol or memorizing a label.

Hazardous Materials Throughout Their Life Cycle

Hazardous-material management does not occur at only one point.

A hazardous material may need to be managed throughout its life cycle:

Acquire / Receive

↓

Store

↓

Use / Handle

↓

Transport

↓

Respond to Releases

↓

Manage Waste

↓

Dispose

Different hazards, exposures, regulations, and management decisions may become important at different stages.

Hazardous-material management means thinking about the material before, during, and after its use.

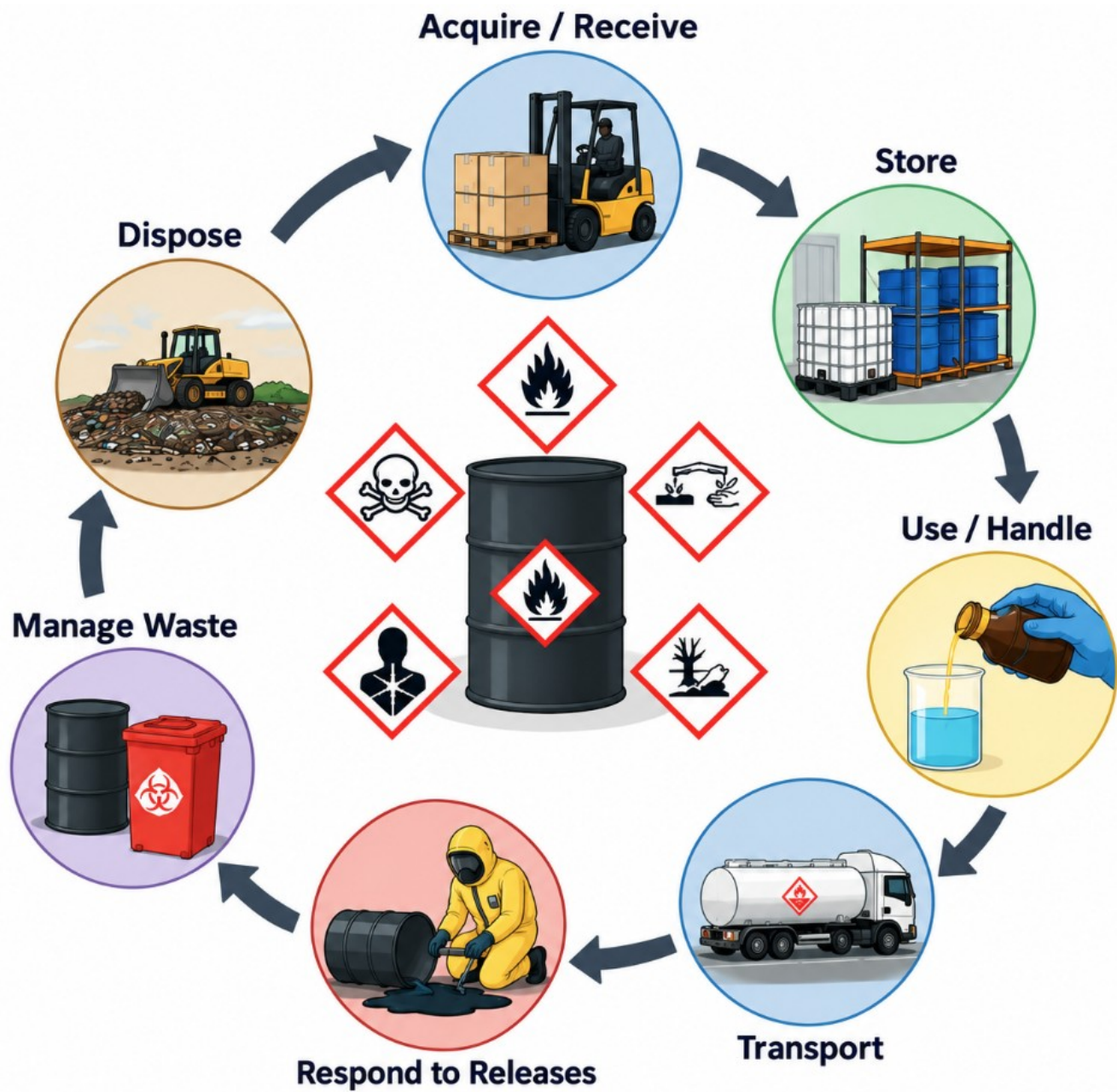


Figure 10: Hazardous materials life cycle from acquisition and receiving through storage, use, transport, release response, waste management, and disposal.

Key Takeaway

Hazardous materials management requires more than identifying whether something is hazardous.

We need to understand:

- what the material is
- what hazardous properties it has

- how people or the environment could be exposed
- how much exposure could occur
- how the material should be handled, controlled, and managed
- what regulations and communication systems apply

Hazardous materials management is the process of recognizing hazards, understanding potential exposures, and making informed decisions about how materials should be managed throughout their life cycle.

Key Terms

Term	Meaning
Hazard	Something with the potential to cause harm.
Hazardous material	A material or substance whose properties can pose a risk to human health, safety, property, or the environment.
Dose-response relationship	The relationship between the amount of exposure and the resulting biological effect.
Hazard communication	Systems used to communicate information about hazardous materials, including labels, Safety Data Sheets, placards, and hazard-identification systems.
Hazardous-material life cycle	The stages through which a hazardous material may be acquired, stored, used, transported, released, managed as waste, and disposed.

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SHM 377: Hazardous Materials Management

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